

CATALAN’S CONSTANT AND THE ARITHMETIC HOLONOMY METHOD: AN OBSTRUCTION

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ABSTRACT. Calegari, Dimitrov and Tang remarked that their arithmetic holonomy method cannot reach Catalan’s constant $G = L(2, \chi_{-4})$ through the known Apéry-type rows “unless some completely new idea is discovered”. We make this quantitative and structural. Using the classification of integral second-order Apéry rows by rational elliptic surfaces, we show that every genus-zero modular host of the χ_{-4} Eisenstein class has its second singularity at distance $\leq \frac{1}{4}$ in the integral coordinate, so the entry condition of the holonomy bound fails by at least 0.077 nats per index even at the uniformisation ceiling; that the p -adic version of the bound cannot repair this because, by Calegari’s theorem that the 2-adic Catalan value is irrational, the conditional function is never 2-adically overconvergent; that Atkin–Lehner quotient hosts place the fold at an orbifold point and carry no conditional function; and that two-host (Hadamard) constructions have a Galois-orbit singular set admitting no pure module. Along the way we exhibit two new unconditional “doubly small” functions on the level-16 host, prove that the number of usable classes is bounded by the number of cusps, and show that the 2-adic divisibility driving the two-row lattice method for G is the Zudilin row’s 2-adic convergence rate to $\zeta_2(2)$ — so both methods stop at the same wall.

1. INTRODUCTION

Let $G = \sum_{k \geq 0} (-1)^k (2k+1)^{-2}$. The arithmetic holonomy method [3, 4] proved $L(2, \chi_{-3}) \notin \mathbb{Q}$ (indeed $1, \pi^2, L(2, \chi_{-3})$ linearly independent) and is, in the authors’ words, blocked for G : on Zagier’s row **E** the relevant inequality reads $e^2 > 16 \cdot \frac{1}{4}$, and on Zudilin’s hypergeometric row $e^4 > \varphi^5$ “by a wide margin” [3, Rem. after Lemma 9.1.4]. This note records what one can prove about that block. The answer has three layers.

Geometry. The χ_{-4} Eisenstein class is realised by integral three-term rows only on one rational elliptic surface (Zagier’s **E**, Kodaira type $I_8 I_2 I_1 I_1$) and its genus-zero cover; all such rows have integer characteristic roots, hence second singularity $|t_2| \leq \frac{1}{4}$; and the uniformisation ceilings of the two hosts are $16 \cdot \frac{1}{4}$ and, after CDT’s symmetrisation, 64, against denominator rates 2 and 4.2355. Entry fails by 0.614 resp. 0.077 nats per index, and failure of entry means the admissible function space is a continuum (§2).

The p -adic place. The multi-place bound rewards functions that are p -adically overconvergent. The function that carries the hypothesis $G \in \mathbb{Q}$ is $H = bB - aA$; its 2-adic coefficients satisfy $h_n \approx (b\xi_2 - a)a_n$ with $\xi_2 = \frac{1}{2}\zeta_2(2)$ the 2-adic Apéry limit, and $\xi_2 \notin \mathbb{Q}$ (Calegari). So H has 2-adic slope 0 whatever a/b is, and one slope-0 function reduces the p -adic gain to a rescaling (§3). The same fact, in the lattice method, is that the rationality-kernel vector is 2-adically the best vector (§6).

Other hosts and more functions. More fold-regular classes are bounded by the number of cusps; the level-16 host carries exactly two unconditional doubly-small classes, new objects but worth +0.4 each against deficits of order 10 (§4). Atkin–Lehner quotients with $4 \mid N$ carry the class rationally only when every involution index is a square, and then place the fold at

an orbifold point, where there is no Eichler period (§5). The Hadamard product of the Zudilin and Nesterenko rows yields a striking unconditional function W , overconvergent by 14 nats with 2-adic content $24n$, but its singular set is one Galois orbit and admits no denominator-free functions, so entry fails by 11 (§5).

Theorem 1.1 (obstruction). *Let (Γ, t, F) be a genus-zero host carrying the χ_{-4} Eisenstein class with an integral Hauptmodul t and an integral weight-one form F , and let $H = bB - aA$ be the conditional function of the hypothesis $G = a/b$. Then for every admissible archimedean template φ and every choice of p -adic radii consistent with the actual coefficients,*

$$\log |\varphi'(0)| + \left(1 - \frac{1}{m}\right) \sum_p \log R_p \leq \tau(\mathbf{b}),$$

*i.e. the entry condition of the holonomy bound fails for every inventory containing H ; on the best host (Zagier **E**, symmetrised) the deficit at the uniformisation ceiling is 0.0766 nats.*

The proof is the union of §§2–5; the hypotheses are those under which the classification of §2 applies. Everything numerical below is exact or certified as indicated; the documents `CATALAN_OBSTRUCTION.md` and its references in the project repository carry the scripts.

2. GEOMETRY: THE HOST IS FORCED

2.1. Hodge depth. $G = L(2, \chi_{-4})$ is the period of the weight-3 Eisenstein class $E = E_3^{\chi_{-4}, 1}$; an Apéry row for it is a pair $(A, B) = (F, F \cdot D^{-2}\Phi)$ with $\Phi = P(V)E$ an oldform combination and D^{-2} a double Eichler integral. Two integrations give denominators of type $[1, \dots, n]^2$, i.e. $\tau = 2$ in every realisation; a single integration would produce $L(1, \cdot)$, i.e. π . (Weight-drop theorem; Theorem B* of the companion paper.)

2.2. Three-term rows. A three-term integral recurrence $(n+1)^2 u_{n+1} = P(n)u_n - Q(n)u_{n-1}$ with $\deg P, \deg Q \leq 2$ is the Picard–Fuchs system of a rational elliptic surface with four singular fibres when its exponent differences are Kodaira-admissible; Herfurtners list and the Möbius invariant a^2/d identify Zagier’s six rows with Beauville’s six surfaces, with **E** on $I_8 I_2 I_1 I_1$ ($a, b, c = 12, 4, 32$, roots 8, 4). Since $a, d \in \mathbb{Z}$ the characteristic roots are algebraic integers, so rational roots are integers with gaps ≥ 1 and $|t_2| = 1/\lambda_2 \leq \frac{1}{4}$ at every placement of the cusp (cusp-move theorem). The only genus-zero cover is level 16 ($X_0(32)$ has genus 1), with roots 4, $2\sqrt{2}$ in the x -coordinate. [Classification: `HERFURTNER_CLASSIFICATION.md`; rigidity: `CUSP_MOVE_PROGRAM.md`, Thm. 4.]

2.3. Ceilings. For the unsymmetrised host $\mathbb{P}^1 \setminus \{0, t_2, \infty\}$ (fold t_1 filled in by the hypothesis) the optimal multivalent template is Kodaira’s, $|\varphi'(0)| = 16|t_2|$ [3, §1], so entry $= \log(16 \cdot \frac{1}{4}) - 2 = -0.614$. CDT’s symmetrisation uses the involution σ of the host and the coordinate $y = x + \sigma(x)$ -type descent. On level 8, $\sigma(x) = x/(4x - 1)$ fixes 0 (cusp) and $\frac{1}{2}$ (free), so the y -line is $\mathbb{P}^1 \setminus \{0, -\frac{1}{8}, \infty\}$ with an order-2 cone point at $y = 1$, of conformal size 64 at 0 once the fold image $-\frac{1}{8}$ is dropped: entry $= \log 64 - \tau = 4.1589 - 4.2355 = -0.0766$ with CDT’s $m = 14$ inventory and $\tau = 16603/3920$. On level 16 the descending involution is $\tau \mapsto \tau + \frac{1}{2}$; both fixed points are cusps, the y -line is the level-8 x -line rescaled by -4 (verified to q^{60}), there is no cone point, and the ceiling is 16: entry -1.463 . [`CATALAN_THREE_PERIOD.md`, appendix.]

2.4. Entry failure is fatal. When $\sum_v \log R_v \leq \tau$, the set of power series with the given denominators and analytic continuation has the cardinality of the continuum [4, §2], [5]; no dimension count can then distinguish a conditional function from the unconditional ones. Hence Theorem 1.1 is a theorem about the method, not about a particular inventory.

3. THE p -ADIC PLACE CANNOT HELP

3.1. The multi-place bound. In the form proved in [8, Thm. 2] (a twist of the slopes-method proof of [3, §6.3]): if the m functions have p -adic radii $R_{p,i}$, the bound reads

$$m(\log |\varphi'(0)| - \tau) + \frac{1}{m} \sum_i \sum_p \log R_{p,i} + (m-2) \sum_p \log R_{p,\min} \leq \text{BC}(\varphi),$$

which for uniform radii is $(m-1) \sum_p \log R_p$ on the left. The graded pieces of the evaluation module are twisted by $R_{\min}^{n-D}$: the p -adic gain is governed by the worst function.

Lemma 3.1. *Let (a_n, b_n) be an Apéry row with $a_n \in \mathbb{Z}$, $\liminf v_p(a_n)/n = 0$, and a p -adic Apéry limit $\xi_p \in \mathbb{Q}_p$ of positive slope: $v_p(b_n - \xi_p a_n) \geq \sigma n - o(n)$, $\sigma > 0$. If $\xi_p \notin \mathbb{Q}$ then for all $a, b \in \mathbb{Z}$, $b \neq 0$, the function $H = \sum (bb_n - aa_n)x^n$ has p -adic slope $\liminf v_p(h_n)/n = 0$.*

Proof. $h_n = b(b_n - \xi_p a_n) + (b\xi_p - a)a_n$ with $b\xi_p - a \neq 0$ of fixed valuation v_0 ; the first term has valuation $\geq \sigma n - o(n)$, so for n with $v_p(a_n) = o(n)$ one has $v_p(h_n) = v_p(a_n) + v_0$. $\square$

For Zagier **E**: $v_2(a_n) = 2s_2(n)$ (zero along powers of 2), $\xi_2 = \frac{1}{2}\zeta_2(2)$ with slope $\sigma_2 = 5$ in x (Theorem F of the companion paper), and $\xi_2 \notin \mathbb{Q}$ by [2]. Measured: H has slope 0 in x and -2 in the symmetrised y (the descent $t = \frac{1}{2}(y - \sqrt{y^2 - y})$ is not 2-integral, so $\varsigma_y = 2\varsigma_x - 2$ with no floor). The pure polylogarithm orbit $\text{Li}_j(4x)$ has slope 2 in x . By scale covariance of the bound under $x \mapsto x/p^v$, the net p -adic contribution of an inventory whose worst function has slope 0 is $\frac{1}{m} \sum_i \varsigma_i \log p$ minus the rescaling needed to make the worst slope nonnegative; for the level-8 inventory this is zero or negative (measured: entry moves from -0.0766 to -0.855 under the measured profile). This proves the p -adic half of Theorem 1.1. The same argument applies at every prime and every placement: $\xi_p = a/b$ would be required, which the hypothesis $G = a/b$ excludes whenever $\xi_p \neq \xi_\infty$ as numbers — and a p -adic number cannot equal a real one except through $\mathbb{Q}$.

4. MORE FUNCTIONS

4.1. Fold-regularity and the cusp count. A class Φ on the host gives a conditional function iff the Eichler period polynomial of Φ at the fold cusp has vanishing constant term (and, when the fold cusp is not the cusp 0 of Theorem B*, also vanishing $\log^2$ -coefficient); then ξ is read off the linear coefficient. One hypothesised linear relation among the periods supplies one generator whose θ -orbit has the size of the holonomic rank; the number of such generators is at most $\#\text{cusps} - 1$. Fold-regularity is a condition of codimension μ , the Jordan block size at the fold, so only rows with simple dominant root qualify. [CDT_NONCONGRUENCE.md §10; HOLONOMY_LINDEP.md.]

4.2. Doubly-small functions at level 16.

Proposition 4.1. *On the level-16 host ($x = \eta_2 \eta_{16}^2 / \eta_1^2 \eta_8$, $F = \eta_2^4 / \eta_4^2$, fold cusp $\frac{1}{2}$) the fold-regular subspace of each orientation of the χ_{-4} Eisenstein space is two-dimensional, and each contains a target-zero class:*

$$\Phi_0^{\text{in}} = (1 + 4V_2 - 32V_4)E_3^{\chi_{-4}, 1}, \quad \Phi_0^{\text{out}} = (1 + 3V_2 - 4V_4)E_3^{1, \chi_{-4}}.$$

Their companions $B_0 \in \mathbb{Q}[[x]]$ have $\xi_\infty = \xi_2 = 0$, archimedean radius $\sqrt{2}/4$ against the host's $\frac{1}{4}$, $d_n^2 B_0 \in \mathbb{Z}$ (sharp), and 2-adic slopes $\frac{3}{2}$ resp. $\rightarrow 1$; they are $\mathbb{Q}(x)$ -independent of each other and of A . The outer period functional is $\xi_\infty^{\text{out}} = \frac{3}{2}\zeta(2)(P(1) - P(2))$, not the Theorem-B* value.

These are unconditional functions small at both places — precisely the objects the method wants, and the first such on any Catalan host — but each is worth about +0.4 of margin and the Eisenstein space supplies two; a cap of [1, §4.3] bounds the number of such functions on any host. [CATALAN_THREE_PERIOD.md, numerically certified at $N = 220$.]

5. OTHER HOSTS

5.1. Atkin–Lehner quotients. For odd weight k , $|_k W_Q$ on $M_k(\Gamma_0(N), \chi_{-4})$ carries a factor $Q^{k/2}$, so eigenvectors are rational iff every Q in the involution group is a square; of the 23 genus-zero groups $\Gamma_0(N) + W$ with $4 \mid N$ this leaves 10. On each proper quotient that carries a row $(\Gamma_0(16) + 16, \Gamma_0(20) + 4, \Gamma_0(36) + 4, 9)$ the nearest singularity is an order-2 orbifold point (an AL fixed point), not a cusp: there is no Eichler period there, the fold-regular space is $\{0\}$, and the Apéry limit is not in $\mathbb{Q} + \mathbb{Q}\zeta(2) + \mathbb{Q}G$ (e.g. on $\Gamma_0(16) + 16$ the fold is at $x = 1/(6 + 4\sqrt{2})$). The best geometry found on any such host is $\Gamma_0(12)$ with roots $(4, 2)$ — +0.69 nats of score over **E** — cancelled exactly by five remaining singular points with no descent (ceiling ≤ 4). $\lambda_2 < 1$ occurs nowhere. [CATALAN_AL_HOSTS.md.]

5.2. The Hadamard host. Let $(a^Z, b^Z), (a^N, b^N)$ be the cleared Zudilin and Nesterenko rows at the aligned indices $(3n, n)$, with $v_2(a_n^Z b_n^N - a_n^N b_n^Z) \geq 24n - O(\log n)$ (the engine of the two-row lattice method). The Hadamard products are holonomic with singular set the pairwise products of the rows' singular points, and

$$W := A_Z \odot B_N - A_N \odot B_Z = A_Z \odot (B_N - GA_N) - A_N \odot (B_Z - GA_Z) \in \mathbb{Q}[[x]]$$

is unconditional, overconvergent by 14.44 nats beyond $A_Z \odot A_N$, of denominator type $[1.6n]^2$, and the unique rational element of its module regular at the fold. Its minimal recurrence has order 4, degree 96, leading polynomial irreducible over $\mathbb{Q}$ with the four product points as roots. Because those points form a single Galois orbit, every rational algebraic or polylogarithmic function on the host branches at all four, including the one W removes: there is no pure module, $\tau^b \geq 10.67$, and entry fails by 11.1 (the 2-adic term is a penalty: $v_2(w_n) = -(2n - 1)$ after the rows' own 2-denominators). Isolating the cost: Zudilin alone -4.2 , 3-sectioned -15.6 , Nesterenko -16.9 , Hadamard -16.9 — the product is a wash; the alignment is the damage. [HADAMARD_HOST.md.]

6. THE SAME WALL IN THE LATTICE METHOD

The two-row lattice method divides integer combinations of the two rows by $D_{6n}^2 2^{\lfloor kn \rfloor}$ and selects a short vector; it reaches worthiness $1 - \varepsilon$ as $k \uparrow k^* = 22.35$ [6, 7], and for $k > k^*$ would prove $G \notin \mathbb{Q}$ if the selected form were nonzero (Lemma P2). Two exact identities locate the obstruction [9, P2_HOLONOMIC.md]:

$$v_2(X_n \xi_2 - Y_n) = v_2(X_n) + 24n - 1 - 4s_2(3n), \quad v_2(X_n U_n - V_n Y_n) = v_2(X_n) + v_2(V_n) + 24n - 1 - 4s_2(3n),$$

so the $24n$ cross-divisibility is the Zudilin row's 2-adic convergence rate to $\xi_2 = \zeta_2(2)$; and the rationality-kernel vector satisfies $\ell_2(c^{\ker}) = (X_n U_n - V_n Y_n)(a - b\zeta_2(2))$ — the vector that is archimedean-worst (form 0 if $G = a/b$) is 2-adically best. The continued fraction of the lattice is G -free; G selects one rung of it at a cost of 14.43 nats per index, and its parity (which is the cone condition) shows no law to $n = 200$. Thus the lattice method and the holonomy method stop at the same statement: *the hypothesis $G \in \mathbb{Q}$ is false at $p = 2$, and nothing 2-adic can be extracted from it.*

7. THE CYCLOTOMIC REALISATION, AND A WIDTH LAW

$G = \text{Ti}_2(1)$ with $\text{Ti}_2(x) = \sum_{k \geq 0} (-1)^k x^{2k+1} / (2k+1)^2$, a function on $\mathbb{P}^1 \setminus \{0, \pm i, \infty\}$ regular at ± 1 with lcm-free denominators n^2 ; this is the $N = 4$ analogue of $L(2, \chi_{-3}) = \frac{2}{\sqrt{3}} \text{Im Li}_2(e^{i\pi/3})$, realised without any Apéry row, as in [3, §9]. The conditional function $H = \int_0^x 2 \text{Ti}_2(t)(1 - t^2)^{-1} dt$ has monodromy $-2\pi i G$ at $x = 1$, so $H + G \log(1 - x^2)$ is regular there under $G \in \mathbb{Q}$ (full descent needs $1, \pi^2, G$ dependent); its type is $n[1..n]^2$, CDT's own. The pure module has ten $\mathbb{Q}(x)$ -independent members, five lcm-free. Yet entry is -0.563 , essentially the unsymmetrised modular value, for a reason that is a theorem: *the λ -template on a degree- d cover of $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ ramified at the base cusp has conformal size exactly $16^{1/d}$* , and $\mathbb{P}^1 \setminus \{0, \pm i, \infty\}$ is the $d = 2$ cover ($u = x^2$). Symmetrisation by $x \mapsto -x$ merges two punctures (doubling τ^b for $\log 4$ of ceiling, net loss), whereas CDT's Atkin–Lehner involution deletes one; the Möbius stabiliser of $\{0, \infty, \pm i\}$ fixing 0 is $\{\pm 1\}$, so no better involution exists; entry decreases with m ; the $\mathbb{Q}(i)$ -coefficient variant is neutral. [CATALAN_MU4.md; calibrated against three of CDT's printed numbers.]

Together with §2 this gives a uniform statement covering every host in this note: *if the fold is a cusp of width w and the second singularity is at t_2 in the integral coordinate, then entry $\leq \frac{1}{w} \log 16 + \log |t_2| - k$, with equality in the thrice-punctured case.* CDT's host: $w = 1, t_2 = 1, k = 2, +0.77$; Zagier **E**: $t_2 = \frac{1}{4}, -0.61$; $X_0(12)$ placement $\{0, \pm 1, \infty\}$: $w = 2, t_2 = 1, -0.61$; μ_4 : $w = 2, t_2 = 1, -0.61$ (realised -0.56 from the richer pure module); level 16: $w = 2, t_2 = \frac{1}{2}, \leq -1.30$ (realised -1.46). For G every term is pinned: $k = 2$ by Hodge depth, and $16^{1/w} |t_2| \leq 4$ on every host constructed.

8. WHAT WOULD EVADE

Theorem 1.1 has three hypotheses a proof must negate: (i) the class is Eisenstein of weight 3 on a modular curve (Hodge depth 2, the Beauville surface); (ii) the host has genus zero with an integral Hauptmodul; (iii) the only arithmetic inputs are archimedean templates and p -adic radii. Negating (i) means realising G as a period of a different motive; the cyclotomic realisation of §7 is the natural candidate and gives the same number by the width law, and conductor-4 hypergeometric realisations carry 16^n denominators, costing exactly $4 \log 2$. Negating (ii) means higher-genus hosts, where the conformal size must be recomputed but the same integrality rigidity is expected. Negating (iii) is the open conceptual question: whether the sign information of the rows (both linear forms are positive moments) can enter a holonomy-type count. We do not know how.

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